ASSESSMENT TOOL-4-PART-2 (CO-4)

Design and Develop Model – Sampling-Based Planning & Data Generation

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Course code & Name: MLA0305- REINFORCEMENT LEARNING

Slot: B

Python code:

```
import numpy as np

import matplotlib.pyplot as plt

np.random.seed(42)

# 1. Generate data using random sampling

N = 200

data = np.random.rand(N, 2) * 10

# 2. Define start and goal

start = np.array([0, 0])

goal = np.array([10, 10])

# 3. Select samples based on distance to goal

dist = np.linalg.norm(data - goal, axis=1)

path_points = data[np.argsort(dist)[:20]]

# 4. Create path

path = np.vstack([start, path_points, goal])

# 5. Calculate path length

path_length = np.sum(np.linalg.norm(np.diff(path, axis=0), axis=1))

# SUMMARY OUTPUT

print("===== SUMMARY OUTPUT =====")
```

```
print("Total Data Points :", N)

print("Selected Samples  :", len(path_points))

print("Path Points       :", len(path))

print("Path Length       :", round(path_length, 2))

print("Planning Status   : SUCCESS")

# 6. Visualization

plt.figure(figsize=(7,6))

plt.scatter(data[:,0], data[:,1], s=15, label="Generated Data")

plt.plot(path[:,0], path[:,1], 'r-', linewidth=2, label="Sampling Path")

plt.scatter(*start, s=100, label="Start")

plt.scatter(*goal, s=100, label="Goal")

plt.title("Sampling-Based Planning & Data Generation")

plt.xlabel("X")

plt.ylabel("Y")

plt.legend()

plt.grid()

plt.show()
```

Summary Output:

```
===== SUMMARY OUTPUT =====
Total Data Points : 200
Selected Samples  : 20
Path Points       : 22
Path Length       : 43.21
Planning Status   : SUCCESS
```

Visualization Output: